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Organization Name	:	Retail And E-commerce Project
Project title	:	AI solution for retail sales forecasting with machine learning.
Github repository link	:	https://github.com/Nyathi240109678/Retail-E-commerce-project#retail-e-commerce-project

DECLARATION FORM

Declaration

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PROJECT DOCUMENTATION

1. Business Background

RetailIQ is an AI solution for the Retail and E-Commerce industry. The project is based on an omnichannel retail business that sells through different sales channels. The business has a large amount of transaction data that includes information about customers, products, sales channels, regions, and dates. This information can be used to understand customer spending and support better business decisions.

The retail business needs to make better use of the historical data that it already collects. Different customers may spend different amounts depending on their customer information, the product category, the sales channel, the region, and the time of the transaction. Looking at these factors together can help the business understand the patterns in its sales.

RetailIQ uses machine learning to turn the available transaction data into an estimate of the expected transaction value. Instead of only looking at past sales manually, the solution learns from historical records and uses the information in the dataset to estimate sales_amount. This allows the retail business to use its existing data in a more useful and data-driven way.

The project is therefore focused on using Artificial Intelligence and Python to solve business problems in Retail and E-Commerce. The solution is designed to provide a simple prediction that can be understood by users without requiring them to perform the calculations manually.

2. Problem Statement

Retail businesses need to understand how much customers are likely to spend in a transaction. Customer spending can be different depending on customer information, product category, sales channel, region, and time. If a business only uses manual calculations or simple averages, it may not get a clear view of customer spending.

The problem is therefore the lack of a simple, data-driven method for estimating transaction value from historical retail data. A large amount of transaction information may be available, but the business still needs a way to use that information to identify patterns and estimate the expected sales amount.

This problem affects the way the business understands its customers and sales. Without a machine learning solution, users may have to depend on manual calculations and basic summaries when trying to understand customer spending. This can make it more difficult to compare actual sales with expected sales and to use the available data effectively.

3. Solution

RetailIQ is an AI-based model designed to estimate the final transaction value (sales amount) at checkout for a customer. The model is trained and tested using historical retail data captured from 2024–2025. The solution uses quantity, unit_price, discount_pct, date, sales_channel, payment method, and region to make the prediction. The solution also considers data leakage by avoiding variables that directly calculate sales_amount, like using engineered features such as gross_sales and discount_amount. This allows the model to learn useful patterns instead of being given the exact mathematical formula for the answer.

4. How will it benefit the community?

The proposed AI solution can benefit surrounding communities by helping local retail and e-commerce businesses make better-informed sales decisions. The model predicts the expected sales amount of a transaction using factors such as quantity, unit price, discount, date, sales channel, payment method, and region. This information can help businesses better understand expected transaction values and make more informed decisions about their sales activities. Stronger and more sustainable local businesses can benefit surrounding communities by continuing to provide goods and services, attracting and retaining customers, and contributing to local economic activity. As local businesses and the economy become more sustainable, increased business activity and competition may also contribute to better access to goods and more affordable prices for consumers. This can help reduce financial hardship and poverty levels within surrounding communities.

Therefore, although the model does not directly provide a community service, its use by local retailers can support more informed business decision-making and contribute indirectly to the economic sustainability and well-being of the surrounding community.

5. BUSINESS OBJECTIVES

5.1 Business Success Criteria

- The model should produce useful predictions with acceptable errors.
- Model performance should be measured using R^2, MAE and RMSE.
- The prediction results should be clear and easy to understand.
- The data preparation and model results should be reproducible.
- The final solution should provide a simple way to estimate a transaction value.

5.2 Requirements

Data

- Historical retail sales data (Dataset).
- Customer information such as quantity, unit_price, discount_pct, date, sales_channel, payment_method and region
- Sales_amount as the target variable.

5.3 Technical requirements

- Python and machine learning tools.
- A trained prediction model.
- Tables or charts to show the results.

5.4 Constraints

- The solution depends on the information available in the dataset.
- The project estimates transaction value and does not predict future stock demand.
- The model cannot make predictions using information that is not available in the dataset.

5.5 Risks

- Poor data quality can affect the results.
- Missing values can reduce the amount of usable data.
- The model may not perform well if the data does not represent future transactions.
- Using variables that directly calculate the target can cause data leakage.

5.6 Sales Regression (Multiple Linear Regression)

Machine Learning Type

- Supervised machine learning
- Regression (predicts a continuous number)
- Algorithm: Multiple Linear Regression

One-Hot Encoding

- Done with pd.get_dummies (X, columns=categorical_features, drop_first=True, dtype=int)
- Each text category becomes its own column of 0 or 1
- drop_first=True avoids the dummy variable trap
- Applied to: sales_channel, payment_method, region
- The prediction tool uses the same encoding, so new data matches the trained model

Method of Converting Words to Numbers

- One-hot encoding for text categories
- Date feature extraction: transaction_date converted with pd.to_datetime(errors="coerce"), then split into year, month, day, day_of_week
- Prediction tool: menu choices (1, 2, 3...) are first mapped to category names using if/elif statements and dictionaries

Features Used

- Numeric: quantity, unit_price, discount_pct, year, month, day, day_of_week
- Categorical: sales_channel, payment_method, region
- Target: sales_amount
- Excluded: gross_sales and discount_amount (calculated from sales data, so they would leak the answer)

Saving and Loading the Model

- Saving (training script):
    - The trained model is saved with joblib.dump(model, "sales_model.pkl")
    - The list of feature column names is saved with joblib. dump(X.columns.tolist(), "model_features.pkl")
    - Both files are saved in the same folder as the program (found with os.path)
- Loading (prediction tool):
    - Checks that both .pkl files exist before running
    - Loads the model with joblib.load("sales_model.pkl")
    - Loads the feature names with joblib.load("model_features.pkl")
    - Uses reindex (columns=model_features, fill_value=0) so the new data has the same columns, in the same order, as training
    - Makes a prediction with model.predict(); no new training happens

5.7 Tools

- Programming
- Python libraries
- pandas (pd)
- numpy (np)
- matplotlib.pyplot (plt)
- scikit-learn (sklearn)

    - train_test_split
    - LinearRegression
    - r2_score
    - mean_absolute_error
    - mean_squared_error
- joblib
- os
- GitHub Developer Platform
- Git
- Python IDE (Coding platform)
- Word
- Excel
- Photoshop
- Grammarly
- Canva
- AI Programs
- PowerPoint
- Retail_sales_dataset.csv
- sales_model.pkl
- model_features.pkl
- Visualization: Actual vs. Predicted scatter plot with a perfect-prediction line

5.8 Techniques

- Data preparation
- Converting non-numerical data into numerical data (one-hot encoding)

- Converting invalid dates to NaN (errors="coerce")
- Removing rows with NaN
- Model training
    - Supervised machine learning approach
    - Use a regression algorithm
    - Linear regression: models the relationship between multiple input features and sales amount using a linear equation
- Train/test split
- Saving the trained model (joblib)
- Loading a saved model (joblib)
- Checking that required files exist before running
- Collecting user input (numbers, date, menu choices)
- Input validation (invalid date, invalid menu choices)
- Matching the new data to the model's features (reindex)
- Predicting with model.predict()

Split Type

- Train/test split (hold-out method) using train_test_split
- 60% training / 40% testing (test_size=0.40)
- Random split with random_state=42 for reproducible results
- Done in the training script only; the prediction tool does no splitting

5.9 Evaluation Metrics

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Actual vs Predicted Sales Amount scatter plot (made with matplotlib)

- Calculated in the training script only; the prediction tool only makes predictions

6. Grammarly Writing Report



